1. Explain the responsibilities of all seven layers of the OSI model.

7. Application	Network services used by applications - HTTP, DNS, SMTP
6. Presentation	Translation, compression and encryption of data
5. Session	Establishes, manages and terminates communication sessions
4. Transport	End-to-end delivery, segmentation, reliability and flow control
3. Network	Logical addressing and routing of packets between networks
2. Data Link	Framing, MAC addressing, local delivery and error detection
1. Physical	Transmits raw bits as electrical, optical or radio signals

2. Compare the OSI model with the TCP/IP model.

Basis	OSI model	TCP/IP model
Layers	Seven layers	Four conceptual layers
Purpose	Reference model for standardization	Practical Internet protocol suite
Upper layers	Application, Presentation, Session	Combined as Application
Lower layers	Data Link and Physical are separate	Combined as Link / Network Access
Development	Model defined before protocols	Protocols existed before the model
Use	Learning, design and troubleshooting	Real-world Internet communication

3. Explain encapsulation and de-encapsulation in computer networks.

ENCAPSULATION AT THE SENDER



Each lower layer adds control information needed for delivery.

DE-ENCAPSULATION AT THE RECEIVER



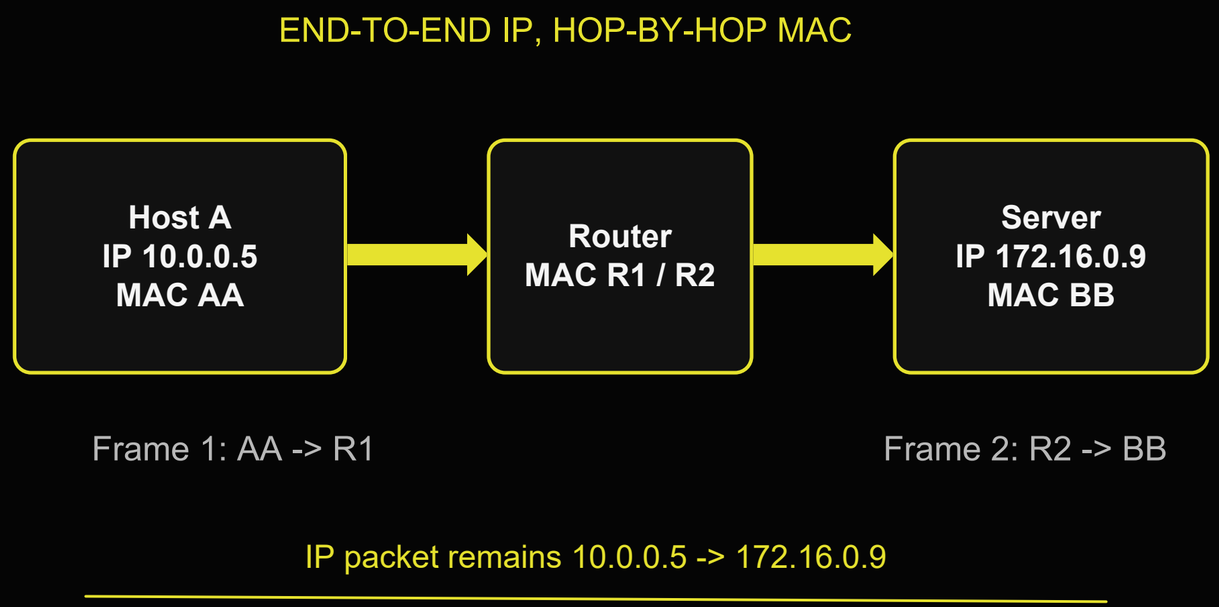
The receiver removes headers in reverse order and delivers the original data.

4. Differentiate between a hub, switch and router based on how they forward data.

Basis	Hub	Switch	Router
OSI layer	Physical	Data Link	Network
Forwarding basis	No address lookup	Destination MAC	Destination IP
Forwarding action	Repeats to every port	Sends to selected port	Selects next network hop
Collision domains	One shared domain	One per switch port	One per interface
Broadcast scope	Forwards all signals	Forwards LAN broadcasts	Separates broadcasts
Main use	Legacy shared Ethernet	Connect devices in a LAN	Connect different networks

5. Why does a device need both a MAC address and an IP address for communication?

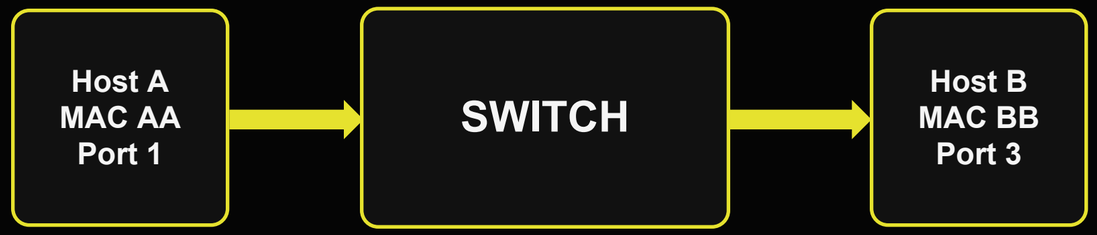
- An IP address identifies the logical source and destination across interconnected networks.
- A MAC address identifies the next network interface on the current local link.
- Routers keep the end-to-end IP addresses but replace the link-layer addresses at every hop.
- ARP resolves the next-hop IPv4 address into the MAC address required to build the local frame.



6. How does a switch learn and use MAC addresses to forward frames?

- The switch records the source MAC address against the frame's incoming port.
- It searches the MAC table for the destination address.
- A known destination is forwarded only through the associated port.
- An unknown destination is flooded through every port except the incoming port.
- Dynamic entries are removed after an ageing period if no frames are seen.

MAC LEARNING AND FORWARDING

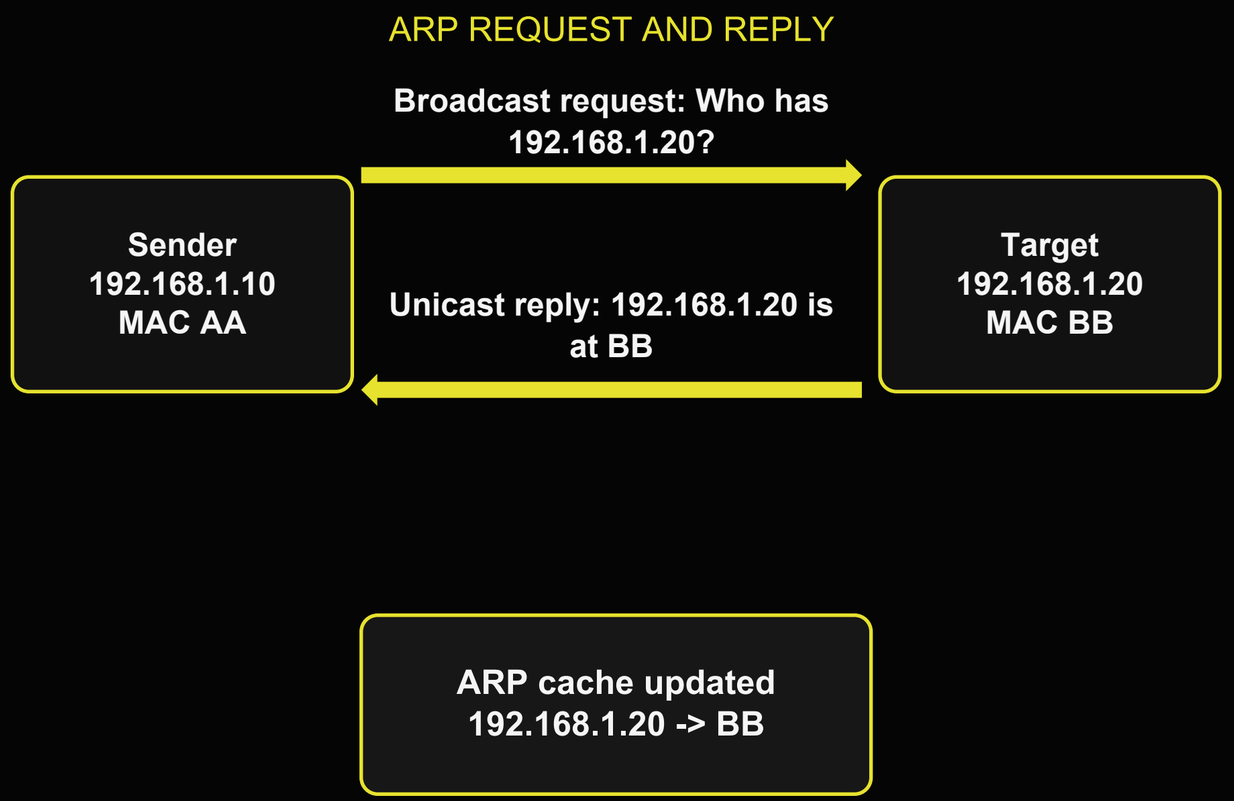


MAC TABLE

AA -> Port 1
BB -> Port 3

7. How does ARP resolve an IP address into a MAC address?

- The sender first checks its ARP cache for an existing mapping.
- If absent, it broadcasts: Who has the required IPv4 address?
- The matching device replies with its MAC address using a unicast ARP reply.
- The sender caches the mapping and uses it to build the Ethernet frame.
- For a remote destination, the sender resolves the default gateway's MAC address.



8. What is a Single Point of Failure (SPOF) in a network?

- A Single Point of Failure is one device, link, or service whose failure can stop the entire network or an important network service.
- Examples include one router, one switch, one Internet link, one firewall, or one DNS server.
- If that single component fails, users may lose Internet access, application access, or communication with other systems.
- It is avoided by adding redundancy, such as backup links, duplicate devices, failover systems, and multiple DNS servers.

Why is it dangerous?

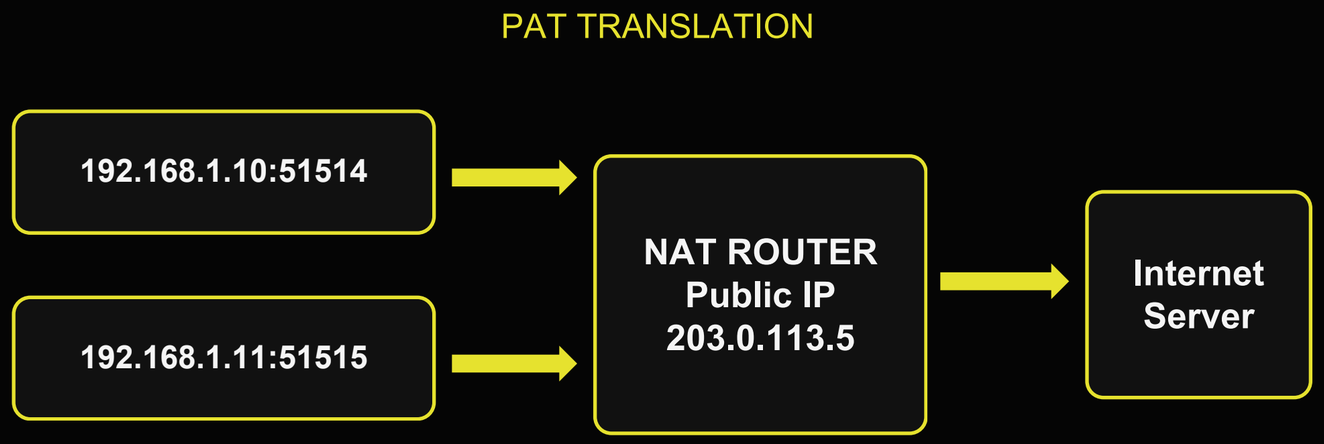
- It can cause complete service downtime without any backup path.
- It increases business loss, user disruption, and recovery time.
- A failure during high-traffic or critical operations can affect many users at once.

9. Compare IPv4 and IPv6.

Basis	IPv4	IPv6
Address size	32 bits	128 bits
Notation	Dotted decimal	Hexadecimal with colons
Header	Variable 20-60 bytes	Fixed 40-byte base header
Broadcast	Supported	Replaced by multicast / anycast
Configuration	Manual or DHCP	SLAAC, DHCPv6 or manual
Fragmentation	Hosts and routers may fragment	Only the source fragments
Address resolution	ARP	Neighbor Discovery using ICMPv6

10. How does NAT allow multiple private devices to access the Internet through one public IP address?

- The router replaces each private source IP with its public IP before forwarding traffic.
- PAT also assigns a unique public-side port to identify every active connection.
- The translation table maps each public port back to the correct private IP and port.
- Reply packets are translated in reverse and delivered to the original device.



TRANSLATION TABLE

192.168.1.10:51514 -> 203.0.113.5:62001
192.168.1.11:51515 -> 203.0.113.5:62002

11. For the classless IP address 192.168.10.70/26, Calculate: Network address, Broadcast address, Number of usable host addresses

/26 means 26 network bits and 6 host bits.

Subnet mask: 255.255.255.192

Block size: 256 – 192 = 64 addresses per subnet.

70 falls in the 64–127 block, so the network starts at 64.

Usable hosts: $2^6 - 2 = 62$ hosts.

Network address

192.168.10.64

Broadcast address

192.168.10.127

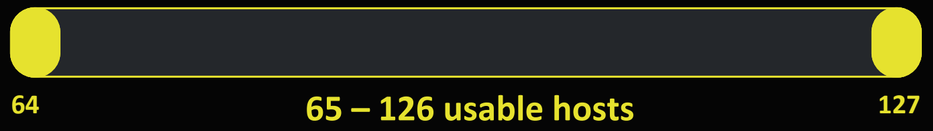
Usable host range

192.168.10.65 – 192.168.10.126

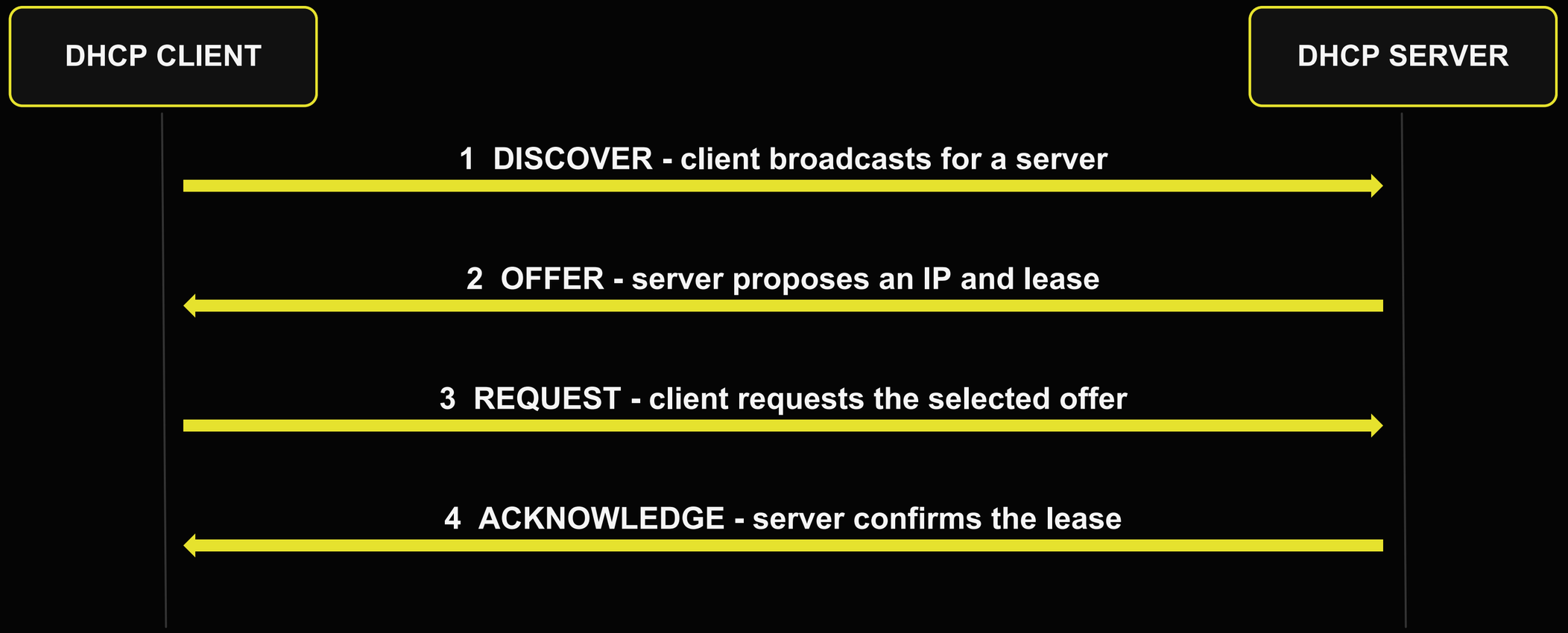
Number of usable hosts

62

Subnet range visual

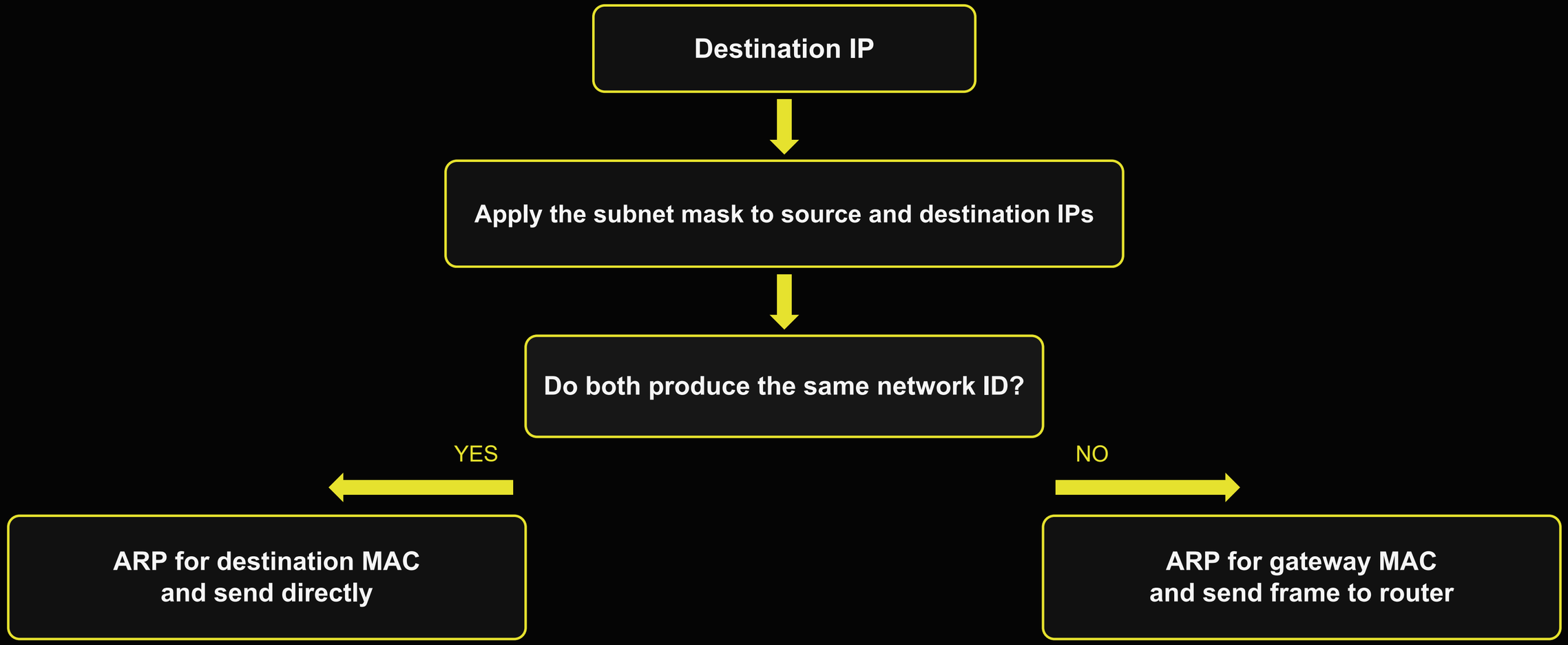


12. Explain the DHCP DORA process.



The acknowledgement includes the IP address, subnet mask, gateway, DNS servers and lease time.

13. How does a host decide whether to send a packet directly or through the default gateway?



14. How does traceroute discover the path followed by a packet?

- Traceroute sends probes with TTL values of 1, 2, 3 and so on.
- Every router decreases TTL by one before forwarding a packet.
- When TTL reaches zero, that router returns an ICMP Time Exceeded message.
- The sender records the router address and round-trip time for each TTL.
- The process ends when the destination responds or the hop limit is reached.

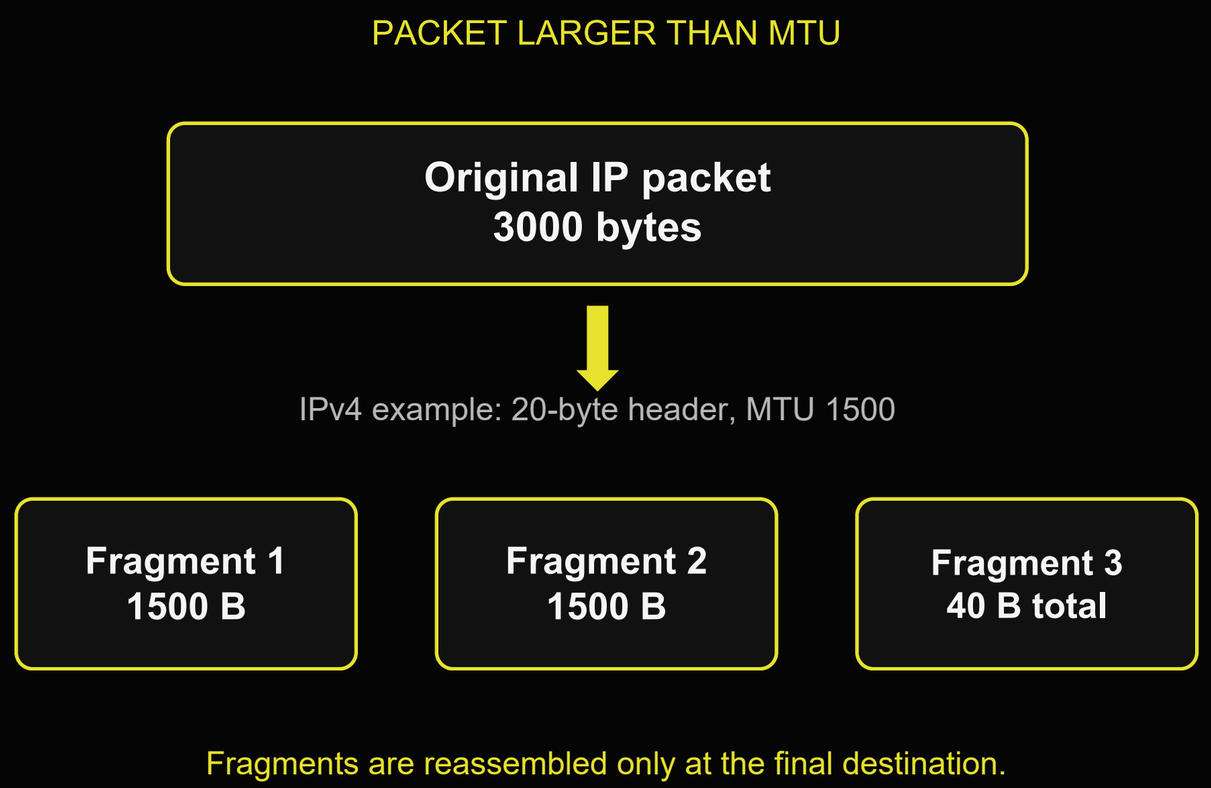
TTL REVEALS ONE HOP AT A TIME



Each expired probe produces an ICMP reply identifying that hop.

15. What happens when an IP packet is larger than the network MTU?

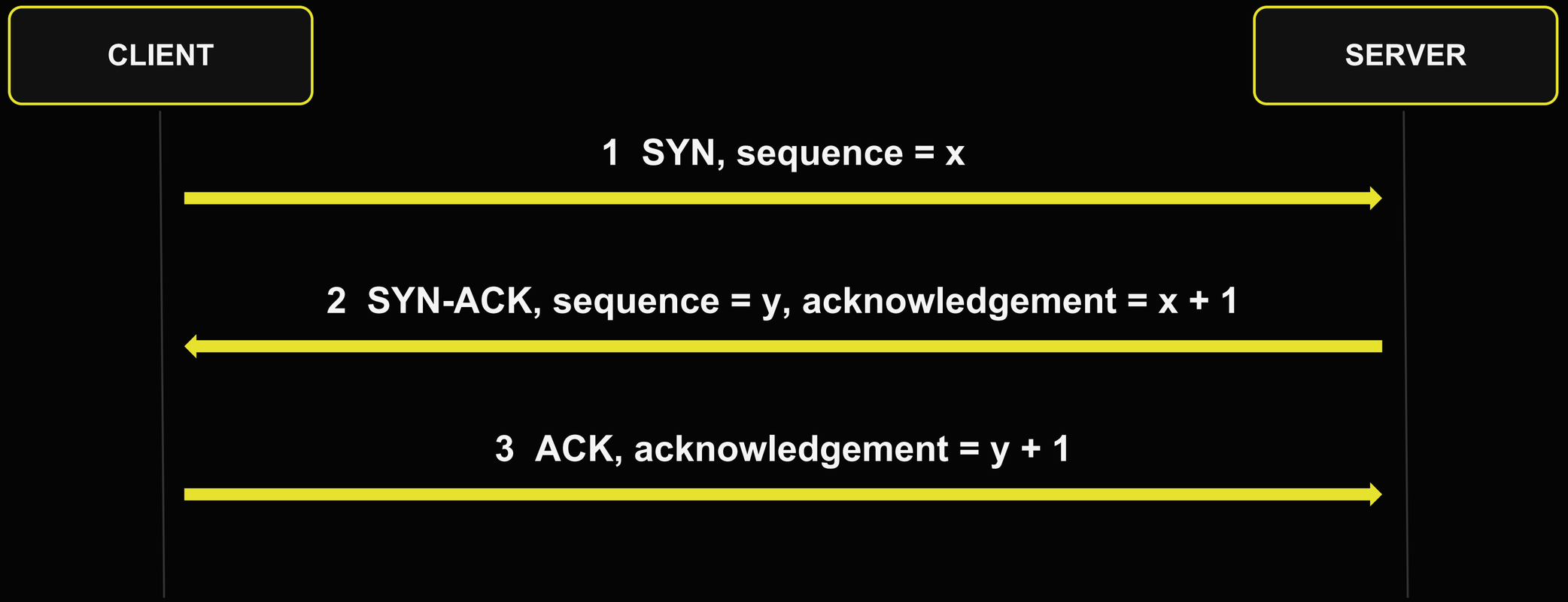
- The MTU is the largest network-layer packet a link can carry without fragmentation.
- In IPv4, a router may fragment the packet when the Don't Fragment flag is clear.
- With Don't Fragment set, the router drops it and returns ICMP Fragmentation Needed.
- IPv6 routers never fragment; the source adjusts packet size using Path MTU Discovery.
- Fragmentation adds headers and makes the entire packet vulnerable if one fragment is lost.



16. Compare TCP and UDP.

Basis	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	ACKs and retransmissions	No delivery guarantee
Ordering	Delivers bytes in order	Datagrams may arrive out of order
Control	Flow and congestion control	No built-in control
Header	Minimum 20 bytes	8 bytes
Best for	Web, email, files, remote login	DNS, voice, gaming, streaming

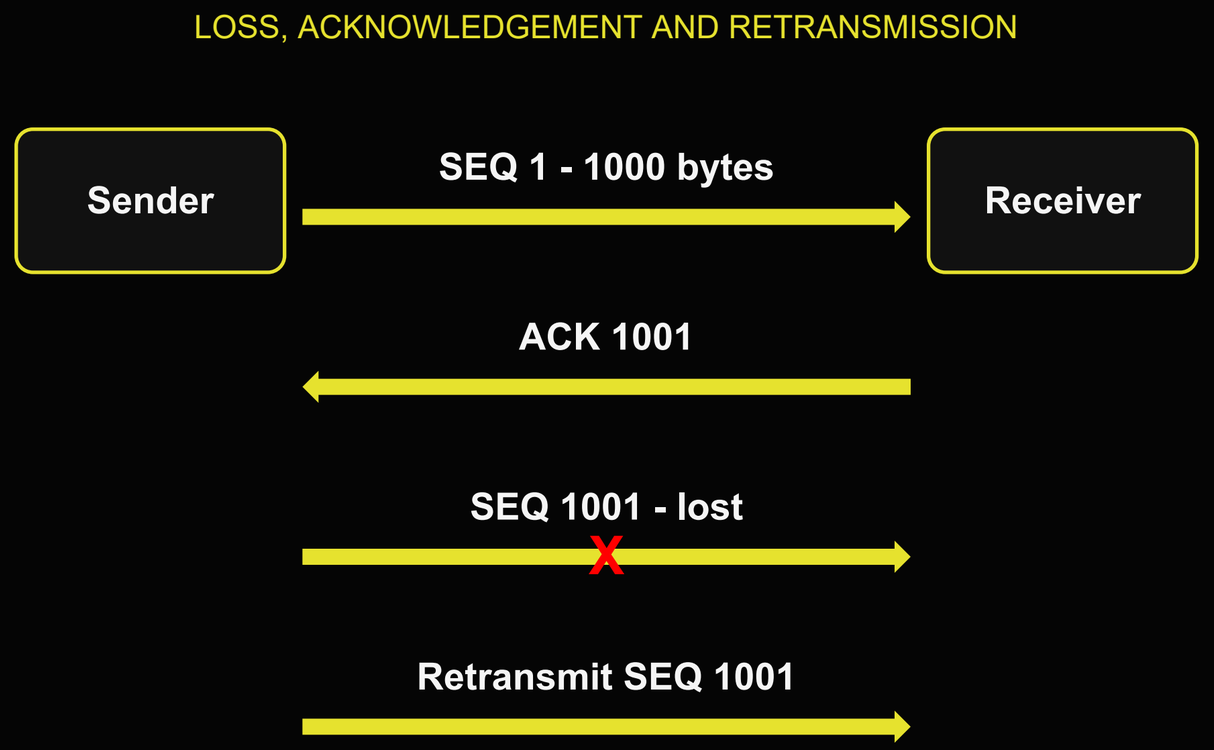
17. Why does TCP use a three-way handshake to establish a connection?



Three messages confirm two-way reachability and synchronize both initial sequence numbers.

18. How does TCP provide reliable and ordered data delivery?

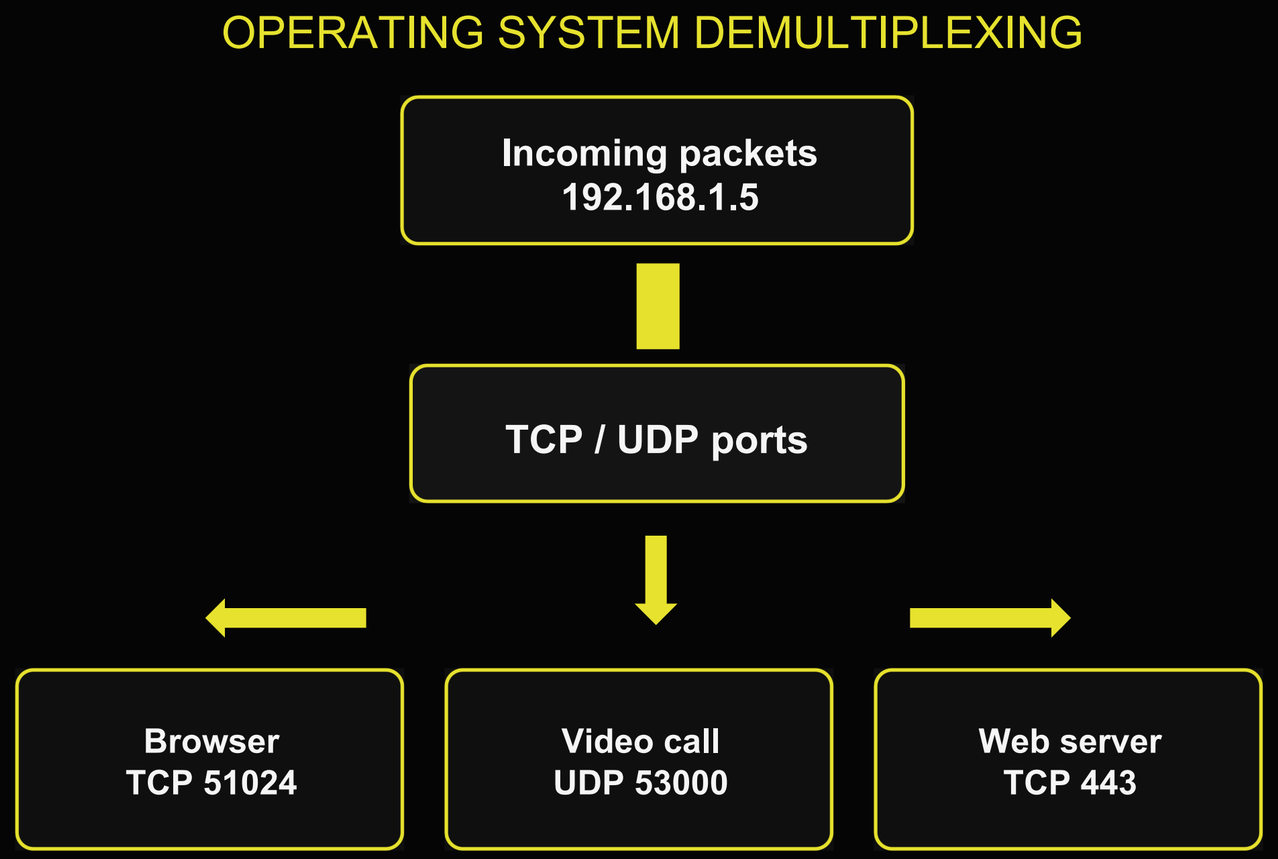
- Sequence numbers identify byte positions and let the receiver reorder data.
- Acknowledgements confirm the next byte the receiver expects.
- A checksum detects corruption in the TCP segment.
- Timeouts and duplicate acknowledgements trigger retransmission of missing data.
- The receive window prevents the sender from overwhelming the receiver.



Only acknowledged bytes move the reliable stream forward.

19. How do ports and sockets allow multiple applications to use the same IP address?

- A port number identifies an application or network service on a host.
- A socket endpoint combines an IP address, transport protocol and port number.
- The operating system uses the packet's protocol and port to deliver it to the correct process.
- A TCP connection is uniquely identified by source IP, source port, destination IP and destination port.
- Therefore, many connections can share one local IP address without mixing their data.



20. How does DNS resolve a domain name into an IP address?

- The browser, operating system and resolver caches are checked first.
- A recursive resolver queries the DNS hierarchy when no cached answer exists.
- The root server directs it to the correct top-level-domain server.
- The TLD server directs it to the domain's authoritative name server.
- The authoritative server returns the record, which is cached for its TTL.

DNS LOOKUP PATH

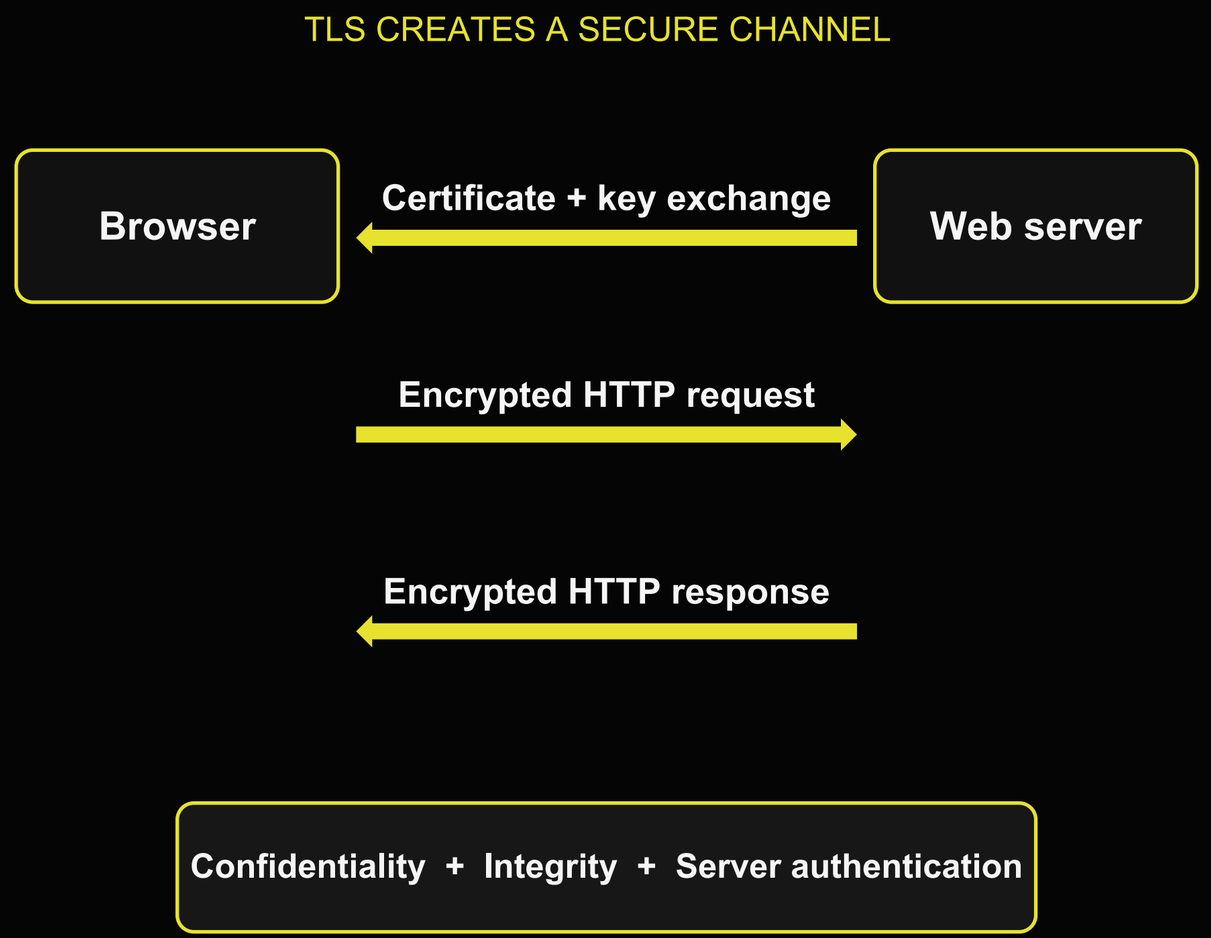


Answer: example.com -> 93.184.216.34

The resolver returns and caches the answer.

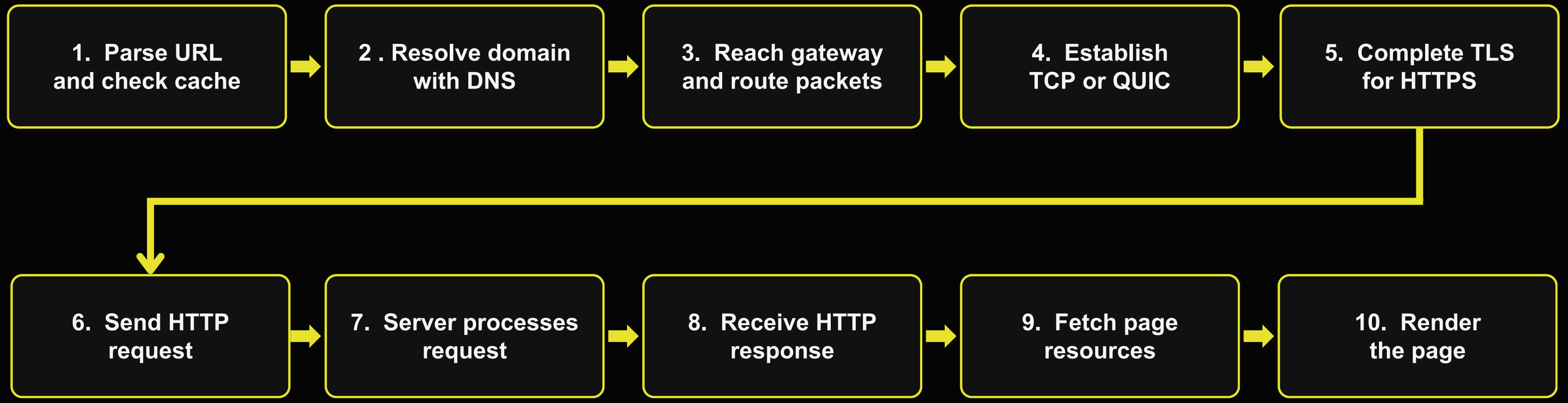
21. How does HTTPS secure communication between a browser and a server?

- The browser starts a TLS handshake after connecting to the server.
- The server presents a digital certificate containing its identity and public key.
- The browser validates the certificate chain, hostname and validity period.
- Both sides establish shared session keys using authenticated key exchange.
- HTTP data is then encrypted and integrity-protected inside the TLS connection.



22. What happens in the network after a user enters a URL in a browser?

FROM URL TO RENDERED PAGE



Caching can skip some steps, while HTTP/3 uses QUIC instead of TCP.

23. **A computer is connected to Wi-Fi but cannot open any website. How would you troubleshoot the problem step by step?**

1 Check Wi-Fi signal and IP configuration

↓

2 Ping the default gateway

↓

3 Ping a public IP such as 8.8.8.8

↓

4 Test DNS with nslookup

↓

5 Check proxy, VPN, firewall and browser

WHAT EACH TEST REVEALS

Result	Likely issue
No valid IP / 169.254.x.x	DHCP or router configuration
Gateway unreachable	Local Wi-Fi, adapter or router
Public IP unreachable	Routing or ISP connectivity
IP works, domain fails	DNS configuration or resolver
Only one browser fails	Browser, proxy, VPN or firewall

24. How would you respond to a suspected network security breach?

A fast, structured response limits damage and preserves evidence.

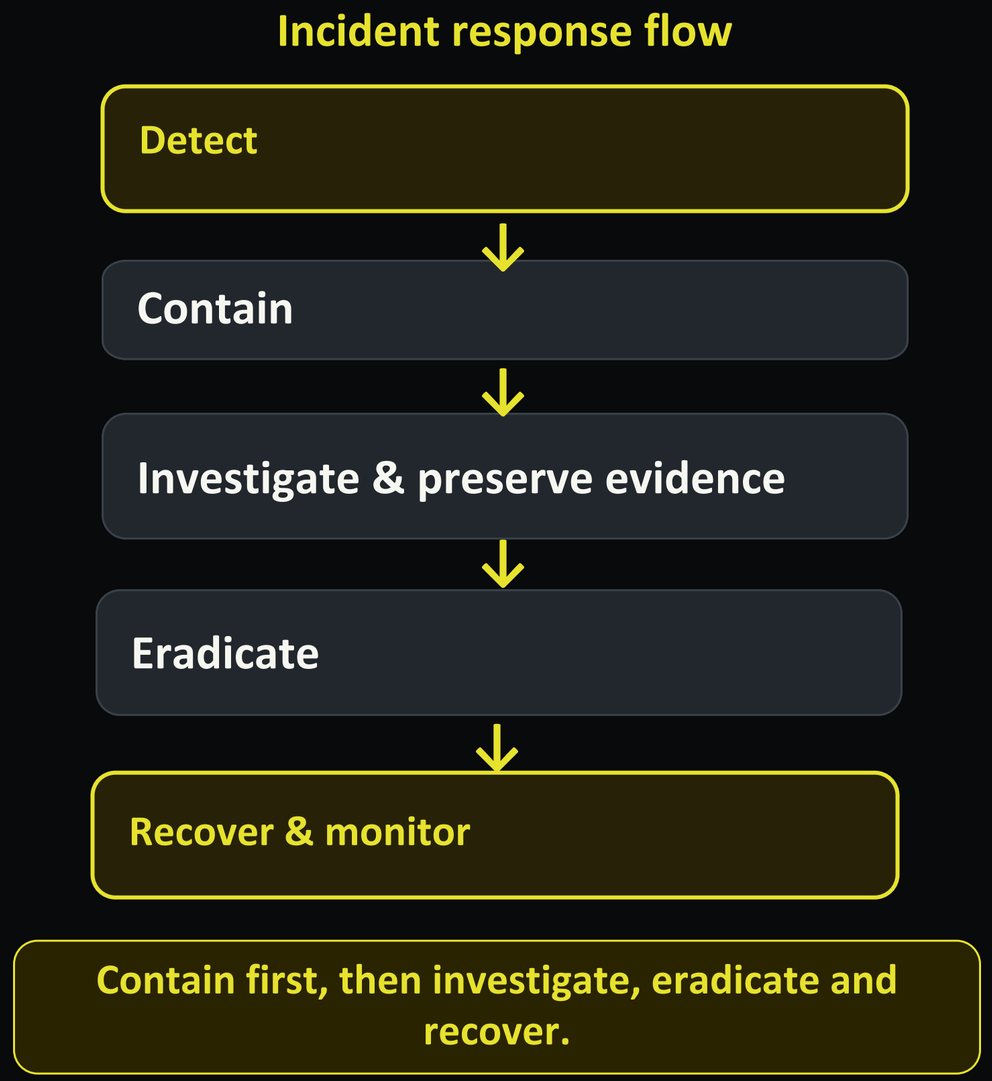
1. Verify the alert using logs, IDS alerts and affected-system checks.

2. Contain the incident: isolate compromised devices or accounts.

3. Preserve evidence before changes—logs, packet captures and timestamps.

4. Remove the cause, patch weaknesses and reset exposed credentials.

5. Recover safely, monitor closely and document the incident.



25. Explain how you would protect an organization against DDoS attacks.

DDoS protection uses multiple layers to absorb, filter and monitor malicious traffic.

1. Use a DDoS protection service or CDN to absorb large traffic floods.
2. Configure firewalls, WAF rules and rate limits for suspicious requests.
3. Use load balancers and redundant servers to spread genuine traffic.
4. Monitor traffic baselines and alert teams when unusual spikes appear.
5. Keep an incident plan with ISP / cloud-provider escalation contacts.

Defense in depth

